UNIT - IV : FIELD TESTING OF SOILS

SPT | PLT | PMT | DMT | VST | CPT | BPT | Geophysical methods

Why field tests? Undisturbed samples are expensive and hard to get in sands, gravels and soft clays. **In-situ tests** measure soil response directly in its natural state, give continuous profiles and let us estimate strength, stiffness and bearing capacity quickly.

1. Classification of field tests

Type	Tests	Gives
Identification	Visual / manual tests	Soil name, consistency
Penetration	SPT, DCPT, SCPT, CPT, BPT	Strength, density (N, qc)
Loading	Plate load test (PLT), pile load test	Bearing capacity, settlement
Expansion / shear	Pressuremeter (PMT), flat dilatometer (DMT), vane (VST)	Modulus, strength, stress state
Geophysical	Seismic refraction, resistivity, cross-hole	Strata boundaries, wave velocity, G
Permeability	Pumping test, Lefranc, packer (Lugeon)	k, groundwater flow

2. Visual identification tests (field identification)

Done on the spot by looking at the sample and doing simple hand tests. Needed for the boring log. (IS:1498 / IS:2720).

Test	Procedure	Result
Dry strength	Crush dried 12-mm pat between fingers	None (silt/sand); low; medium; high (CH clay)
Dilatancy (shaking) test	Wet pat of soil in palm, shake & squeeze	Rapid water appears = silt/fine sand; slow/none = clay

Test	Procedure	Result
Toughness (thread) test	Roll a thread to 3 mm, fold, re-roll near plastic limit	Tough thread = high plasticity clay; weak = silt
Dispersion test	Drop soil in water cylinder, observe settling	Sand settles in 30 s, silt in 30 min, clay stays cloudy
Shine / smear test	Cut with knife	Shiny surface -> clay
Feel & odour	Rub between fingers; smell	Gritty = sand; smooth = silt; sticky = clay; organic odour = peat
Sedimentation, bite test	Gritty on teeth = silt/sand	Quick sizing

Consistency tests for clay: thumb pressed easily - soft; moulded by strong pressure - medium; thumb dents with difficulty - stiff; thumbnail can mark - very stiff; hard - nail cannot mark.

3. Standard Penetration Test (SPT) - IS:2131-1981

The most common test in India. A split-spoon sampler is driven into the soil at the bottom of a borehole by a 63.5 kg hammer falling 750 mm. The number of blows for the last 300 mm of the 450 mm drive is the N value.

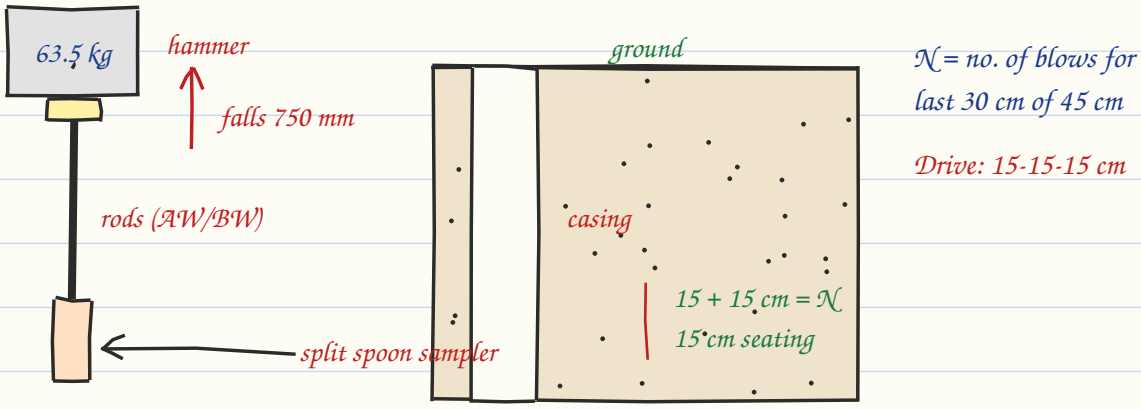


Fig 4.1 Standard penetration test

3.1 Procedure

- Advance borehole to test depth (usually every 1.5 m or change of strata); clean the hole; keep water in hole higher than W.T.
- Lower the split spoon (50.8 OD x 35 ID) on rods; seating drive of 150 mm - record blows.
- Drive next two 150 mm increments - add blows = N. Stop if 50 blows in any 150 mm increment -> refusal (note penetration).
- Pull out the spoon, open it, describe & preserve the sample in jar (SPT sample = disturbed representative).
- Record N for each depth in the log.

3.2 Corrections to N (Peck, Hanson & Thornburn)

Correction	Expression	Remarks
Overburden (CN)	CN = 0.77 log10 (2000/s0') [s0' in kPa], CN <= 2	For sands; N1 = CN x N
Energy / hammer	N60 = N x (ER/60)	Donut hammer ER ~ 45 %, safety 60 %, automatic 80 %
Rod length	Cr < 1 for short rod (< 10 m): 0.75 (3-4 m), 0.85 (4-6 m), 0.95 (6-10 m)	Energy loss in rods
Borehole diameter	Cb = 1.0 (65-115 mm), 1.05 (150 mm), 1.15 (200 mm)	
Sampler liner	Cs = 1.0 (with liner), 1.2 (without)	
Dilatancy (water)	N = 15 + 0.5 (N - 15) for fine/silty sands below W.T. with N > 15	Terzaghi-Peck

3.3 Correlations (Terzaghi & Peck)

Sand N	Density	phi (deg)	Clay N	Consistency	qu (kPa)
0 - 4	Very loose	< 29	< 2	Very soft	< 25
4 - 10	Loose	29-30	2 - 4	Soft	25 - 50
10 - 30	Medium	30-36	4 - 8	Medium	50 - 100
30 - 50	Dense	36-41	8 - 15	Stiff	100 - 200

Sand N	Density	phi (deg)	Clay N	Consistency	qu (kPa)
> 50	Very dense	> 41	15 - 30	Very stiff	200 - 400
			> 30	Hard	> 400

- Clay: cu = qu/2 ~ 6.25 N kPa (Terzaghi) or cu = 6 N roughly.
- Allowable bearing pressure of sand from N via Meyerhof / IS:8009 charts (settlement 25 mm).
- Used also for liquefaction screening, pile capacity (end bearing 40 N60 x Db/B kPa).

Limitations: crude, disturbed sample, depends on hammer energy, operator and boring method; unreliable in gravels (oversize), soft clays (low N), and fine sand below W.T. without care.

4. Dynamic cone penetration test (DCPT) - IS:4968

A cone (50 mm dia, 60 deg) is driven by 65 kg hammer (falling 750 mm) without boring. Blows per 300 mm (Ncd) are recorded. Bentonite slurry can be used for friction reduction in the 100-mm type with casing. N value ~ 1.5 Ncd for fine sand; very cheap for quick checking of the foundation of low-rise structures.

5. Plate Load Test (PLT) - IS:1888-1982

Direct test of bearing capacity and settlement: a rigid steel plate placed at foundation level is loaded in increments and settlement measured. Reaction by kentledge (dead load) or anchors / loaded truck.

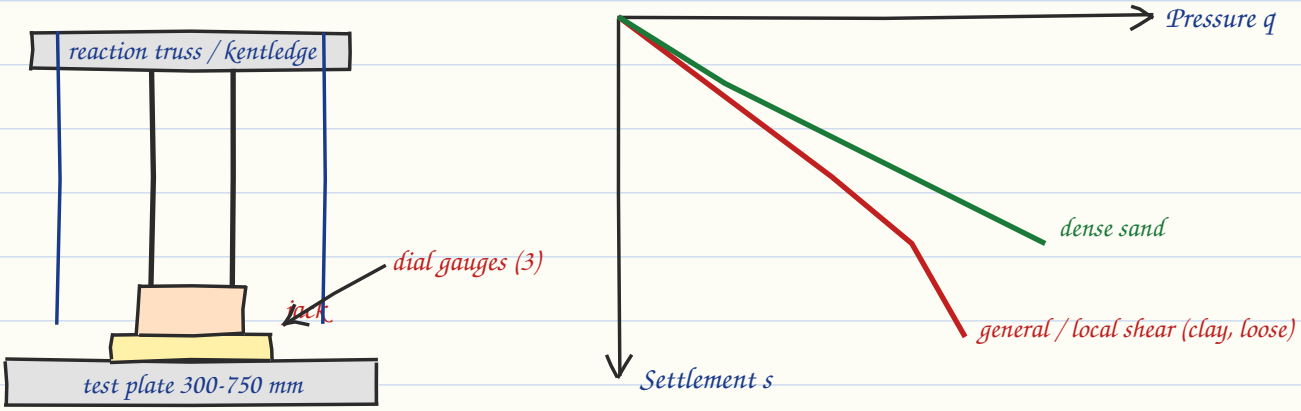


Fig 4.2 Plate load test and typical load-settlement curves

5.1 Procedure

- Excavate a pit of width not less than 5 x plate size, to proposed foundation level; level & clean bottom (fine sand bed).
- Plates: square 300, 450, 600 mm or circular 300-750 mm dia, thickness 25 mm.
- Apply load in increments of 1/5 of the estimated safe load (or 1/10 of ultimate); each stage until settlement rate < 0.02 mm/min (or 1 h) ; settlement by 2 - 4 dial gauges (0.01 mm).
- Continue till failure or 2.5 x design load, or settlement 25 mm (sand) / 10 % of plate width.
- Plot load-settlement curve (log-log or arithmetic).

5.2 Interpretation (size effect)

Soil	Ultimate bearing capacity	Settlement of foundation (Sf) from plate (Sp)
Clay	qf = qp (independent of size)	Sf = Sp x (Bf/Bp)
Sand	qf = qp x (Bf/Bp)	Sf = Sp x [Bf (Bp + 0.3) / (Bp (Bf + 0.3))]^2 (B in m)
c-phi soil	Use equations from c and phi of the soil	-

Example: 0.3 m plate in sand fails at 280 kN/m2; footing width 2 m. qf = 280 x 2/0.3 = 1867 kN/m2 (too optimistic; take lower) ; settlement for plate 12 mm at 100 kN/m2 -> foundation: 12 x [2 x 0.6/(0.3 x 2.3)]^2 = 12 x (1.739)^2 = 36.3 mm.

- Limitations: reflects only soil within 1.5 - 2 plate widths below plate; scale effect, for consolidation of deep clays use the consolidation test; time-consuming/expensive.
- Plate bearing test for pavements uses 300-750 mm plate to find modulus of subgrade reaction k = q/delta (kN/m3).

6. Pressuremeter test (PMT) - Menard

A cylindrical probe (flexible rubber membrane) is lowered in a borehole and expanded radially by pressurised water/gas; volume change vs pressure is measured. Gives in-situ stress-strain behaviour.

Parameter	Meaning / formula
p0	At-rest horizontal pressure (start of linear part)
pf	Creep pressure (end of pseudo-elastic part)
pL	Limit pressure (volume doubles) - used for bearing capacity: qu = k (pL - p0) + sv0
EM	Pressuremeter modulus EM = 2 (1 + n)(V0 + Vm)(Dp/DV)
Cu (clay)	cu = (pL - p0)/Np, Np ~ 5.5

- Advantages: modulus & strength in all soils and weak rock; good for deep foundations and settlement.
- Types: Menard (pre-bored), Self-boring pressuremeter (SBPM) - minimal disturbance, High-pressure dilatometer for rock.

7. Flat Dilatometer Test (DMT) - Marchetti

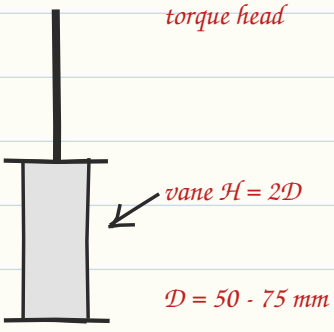
A steel blade (14 mm thick, 95 mm wide) with a thin circular steel membrane (60 mm) on one face is pushed (or driven) into soil at 20 mm/s. At each 200 mm depth, gas pressure inflates the membrane. Two readings: A (lift-off pressure, membrane begins to move) and B (pressure at 1.1 mm expansion).

Index	Formula	Gives
Material index ID	(p1 - p0)/(p0 - u0)	Soil type: <0.6 clay, 0.6-1.8 silt, >1.8 sand
Horizontal stress index KD	(p0 - u0)/sv0'	OCR, K0 (K0 ~ (KD/1.5)^0.47 - 0.6)
Dilatometer modulus ED	34.7 (p1 - p0)	Constrained modulus M = RM ED
Undrained strength cu	0.22 sv0' (0.5 KD)^1.25	Clays

p0 = 1.05 (A - ZM + DA) - 0.05 (B - ZM - DB) ; p1 = B - ZM - DB. Fast (about 1 hour / 20 m), repeatable, cheap; widely used for settlement and liquefaction.

8. Vane Shear Test (VST) - IS:4434

The best in-situ test for undrained shear strength (cu) of soft, sensitive clays and silts. A four-bladed vane (H/D = 2) is pushed into the soil and torque applied slowly; torque at failure T gives the strength of the cylindrical surface.



Su = T / (pi D2 (H/2 + D/6))

T = max torque at failure (N.m)

Rate: 6 - 12 deg per min

Remoulded: rotate 10 turns, retest

Sensitivity St = Su / Sur

Only for soft & medium clays (Su < 100 kPa)

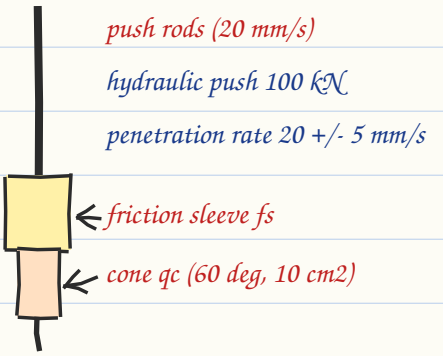
Fig 4.3 Vane shear test

Formula: T = cu (pi D2 H/2 + pi D3/6) -> cu = T / [pi D2 (H/2 + D/6)]. For H = 2D: cu = 6T/(7 pi D3). Example: D = 75 mm, H = 150 mm, T = 62 N.m -> cu = 62 / [pi x 0.075^2 x (0.075 + 0.0125)] = 62 / (0.017671 x 0.0875) = 40 kPa (soft clay).

- Bjerrum correction (clays with Ip high): cu(design) = m x cu(vane), m = 1.7 - 0.54 log Ip (decreasing with plasticity).
- Steps: push vane 0.5 m below borehole bottom; rotate 6 - 12 deg/min; record T vs angle; peak = undisturbed; rotate 10 turns -> remoulded; test again -> St.
- Not suited to stiff clays, sands, gravels or fissured clays.

9. Cone Penetration Test (CPT / Dutch cone) - IS:4968 Pt3

A standard 60-degree cone of 10 cm2 base area is hydraulically pushed at constant 20 mm/s into the soil. Measures cone resistance qc and sleeve friction fs (mechanical or electric cone). With a pore-pressure transducer = piezocone (CPTu).



Outputs:

qc cone resistance (MPa)
fs sleeve friction (kPa)
Rf = fs/qc x 100 (friction ratio)
u2 pore pressure (CPTu)
Sand: high qc, low Rf (< 1 %)
Clay: low qc, high Rf (3 - 6 %)

Fig 4.4 Cone penetration test

Soil	qc (MPa)	Rf (%)	Remarks
Soft clay	0.2 - 1	3 - 6	Higher Rf
Stiff clay	1 - 4	2 - 4	
Loose sand	2 - 5	< 1	
Dense sand	10 - 30	0.5 - 1.5	
Gravel	> 30	< 1	Refusal common

- Correlations: N = qc/ (0.4 MPa to 0.6 MPa) (qc in kg/cm2 = 4 N for sand, 2 N for clay-sand); cu = (qc - sv0)/Nk, Nk = 15 - 20; Es = a qc (a = 2 - 4 sand).
- Bearing capacity of piles (end resistance qb ~ qc), liquefaction analysis, soil classification (Robertson chart, Ic).
- Advantages: continuous profile, fast, repeatable, no sample/hole, cheap. Limitations: no samples; refusal in gravel; need calibration with boreholes.

10. Becker Penetration Test (BPT)

Used in gravelly soils/ cobbles where SPT fails (oversized particles). A double-walled 168 mm closed-end casing (Becker drill) is driven continuously by a diesel pile hammer (rate 55-60 blows/min); BPT = no. of blows for each 300 mm advance. BPT values are converted to equivalent SPT N by Harder & Seed correlations using bounce-chamber pressure to correct for hammer energy:

- Corrected N-equivalent = f(Nb, bounce chamber pressure) ; for liquefaction assessment of gravelly sands/dams.
- Pros: penetrates gravels, up to 150 m; Cons: complicated corrections, no sample in closed type (open-ended Becker allows reverse circulation sampling).

11. Geophysical methods of exploration

Indirect, non-destructive methods using physical properties (velocity, resistivity) of ground. Cover large areas rapidly at low cost; must be calibrated by boreholes.

11.1 Seismic refraction method

A shock (sledgehammer / explosive) generates P-waves; geophones record first arrival times. Waves refracted at the interface with a faster layer (V2 > V1) return to surface and arrive earlier than direct wave beyond the critical distance (Xc).

Critical angle ic = sin-1 (V1/V2)

Depth z = (Xc/2) sqrt((V2-V1)/(V2+V1))

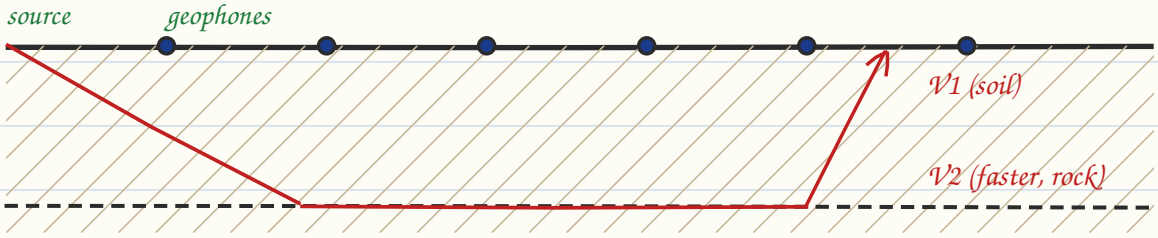


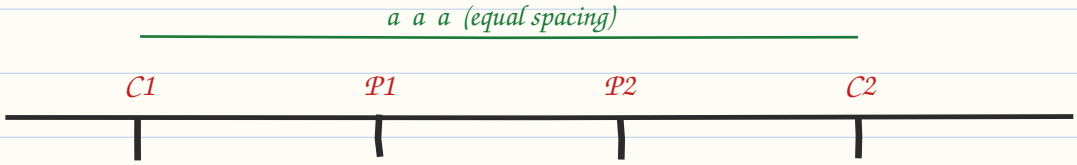
Fig 4.5 Seismic refraction

- Plot time-distance graph; two straight segments; slope = 1/V. Intercept time Ti gives depth: z = Ti V1 V2 / (2 sqrt(V2^2 - V1^2)).
- Typical P-wave velocities (m/s): dry sand 200-1000; wet sand 1000-2000; clay 1000-2500; water 1500; weathered rock 2000-4000; hard rock 4000-6000.
- Limitations: V must increase with depth (hidden layer / velocity reversal not detected).

11.2 Electrical resistivity method

Current is passed through ground by two current electrodes; the potential drop across two potential electrodes gives apparent resistivity. Resistivity depends on moisture, salt content and porosity: dry rock very high, clay & wet soil low.

Current I through C1-C2, potential V across P1-P2:



rho = 2 pi a (V/I) (a = electrode spacing)

Schlumberger: rho = pi (L2 - l2)/(2 l) x V/I, L = AB/2, l = MN/2

Fig 4.6 Wenner and Schlumberger arrays

Material	Resistivity (ohm-m)
Clay, saturated	1 - 100
Wet sand / gravel	100 - 1000
Dry sand	1000 - 5000
Sea water	0.2
Granite, massive rock	10^3 - 10^6

- *Sounding (VES):* electrodes expanded about a centre -> depth increases; *profiling:* electrodes shifted along line at fixed spacing - for lateral variation, buried channel.
- Uses: depth of W.T., bed rock, clay lenses, corrosion & earthing design, location of cavities; limited by electrical interference.

11.3 Cross-hole test

Two or three boreholes at 3 - 5 m spacing; a seismic source (hammer/ SH-wave source) in one hole, receivers (geophones) at same depth in other holes - measures shear (S) wave velocity Vs directly at each depth. (ASTM D4428). Holes are cased with PVC, grouted, inclinometer survey to know exact spacing.

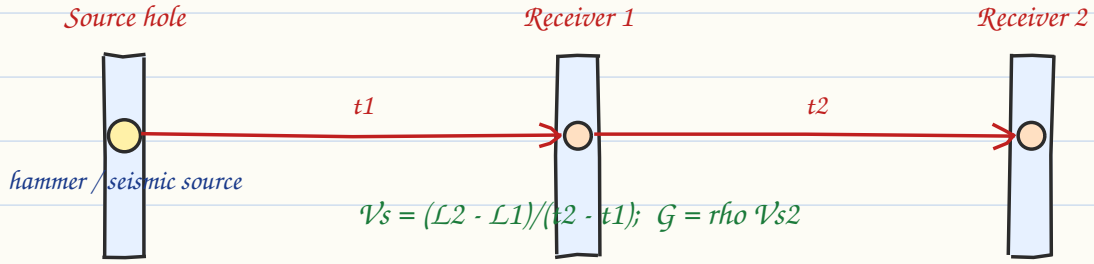


Fig 4.7 Cross-hole test

Calculations: Vs = (L2 - L1)/(t2 - t1) ; Gmax = r Vs ^ 2 ; Emax = 2 (1 + n) Gmax. Used for dynamic analysis, machine foundations, earthquake site response. Down-hole and up-hole tests are cheaper variants using one borehole.

12. Analysis of test results - summary

Test	Result used for	Typical design use
SPT	N60, phi, cu, density	Bearing capacity, pile design, liquefaction
PLT	q-s curve, k	Safe bearing pressure of shallow foundation
PMT	pL, EM	Foundation capacity, deformation
DMT	ID, KD, ED	OCR, K0, settlement
VST	cu, sensitivity	Soft clay embankments, stability

Test	Result used for	Typical design use
CPT	qc, fs, u	Soil profile, piles, liquefaction
BPT	Becker N	Gravel / liquefaction
Seismic refraction	V, depth of rock	Rippability, rock head
Resistivity	rho	W.T., rock, corrosion
Cross-hole	Vs, Gmax	Dynamic soil properties

13. Important questions

- Explain the SPT procedure and corrections applied to N values.
- Describe plate load test. How do you find the bearing capacity & settlement of footing from it?
- Write short notes on PMT and DMT.
- Explain the vane shear test and its limitations.
- What is CPT? List its advantages over SPT.
- Write about Becker penetration test.
- Explain seismic refraction and electrical resistivity methods for subsurface exploration.
- Describe cross-hole test and how shear wave velocity is determined.